Visual Memory for Future Prediction

Research question

When a model can keep only a limited memory, does what it remembers help it predict what comes next? We test two ideas: earlier visual evidence may remain useful, and some observations may be safer to forget than others. This study examines those ideas before learning an automatic memory selector.

Experimental structure

We use 78 Assembly101 videos: 58 for training and 20 for validation, with different people in the two sets. A fixed I-JEPA encoder turns frames into visual features, and a shared predictor learns future features. Each predictor is trained with the same four memory conditions. We repeat training three times from different initializations.

A Test whether history helps

Compare the same predictor with recent observations only, FIFO history (forget the oldest), Reservoir history (sample from the past), and all available history. The recent buffer holds 8 observations; bounded history holds 16. Keeping the predictor and future targets fixed isolates the effect of the supplied memory.

B Test whether the deletion choice matters

From the same full memory, try all 17 possible deletions and evaluate both the immediate prediction and the next 32 observations. Every choice receives the same future stream and random draws. This tests whether a better deletion can preserve useful information, rather than benefiting from an easier continuation.

C Test continuous irreversible forgetting

Replay the videos with FIFO, Reservoir and Uniform random deletion. Once an observation is deleted, it cannot be recovered. This tests how fixed policies behave under a persistent memory limit, rather than in a single isolated deletion.

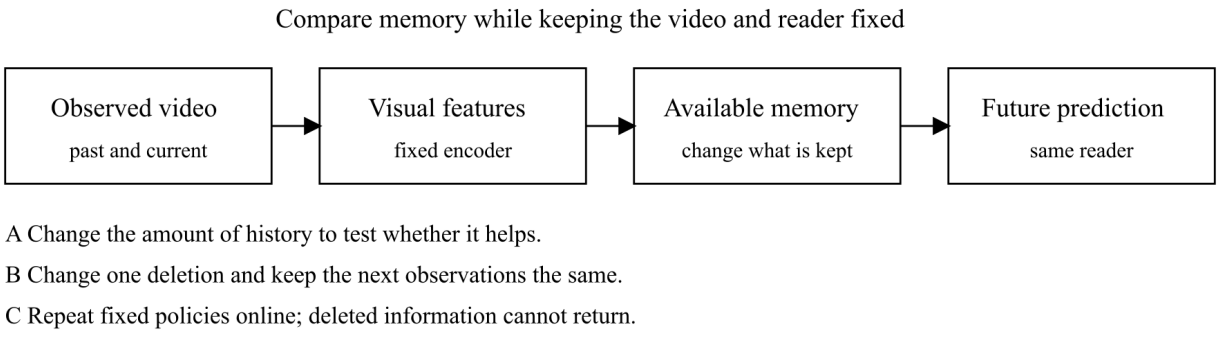


Figure 1. The shared prediction pipeline and the purpose of each experiment. The video and predictor are held fixed when comparing memory conditions.

Future prediction results

We compare predicted and actual future visual features using normalized mean squared error (MSE); lower is better. Gain means baseline error minus candidate error. Scores give equal weight to validation participants, then average the three training runs. Confidence intervals use 10,000 participant bootstrap resamples of the 10 validation people; an interval containing zero leaves the direction uncertain.

Table 1. Prediction MSE at four future horizons. All memory conditions use the same predictor within each training run.

Memory	**0.5 s**	**2 s**	**8 s**	**32 s**
Recent	0.16278	0.29637	0.40232	0.47849
FIFO	0.16341	0.29610	0.39907	0.47453
Reservoir	0.16285	0.29576	0.39987	0.47548
Full	0.16380	0.29713	0.40001	0.47553

Full history reduces the mean error relative to Recent by 0.574% at 8 seconds and 0.619% at 32 seconds. Its shorter-horizon mean errors are slightly higher. FIFO has the lowest mean error at the two longer horizons; this ranking is descriptive because policy-to-policy confidence intervals were not reported.

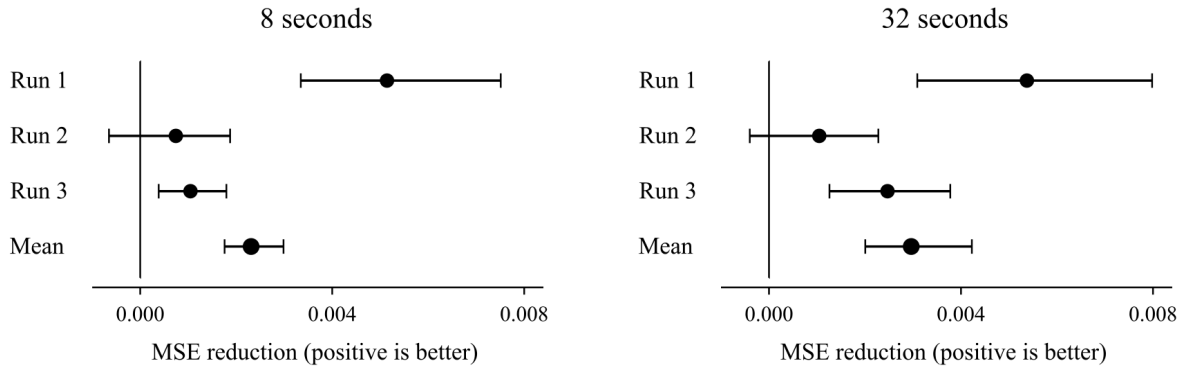


Figure 2. Full history gain over Recent at 8 and 32 seconds. Dots are MSE reductions and bars are 95% confidence intervals. Mean averages the runs within each participant, so the sample remains 10 people.

The positive pooled intervals support a longer-horizon history benefit. Run 2 remains uncertain at both horizons, showing that the benefit is not equally stable across training runs. Predicting visual features well does not by itself establish correct action understanding.

Deletion choice and online memory

For each deletion, the score averages immediate MSE and the mean MSE over the next 32 observations, with equal weight on both and on the four horizons. The best deletion is selected once using this joint score. Table 2 reports the cost reduction relative to the fixed deletion baselines.

Table 2. Mean gain from the best available deletion. Source is the policy that formed the starting memory; baseline is the deletion used for comparison.

Source	Baseline	Joint MSE gain	95% CI
FIFO	FIFO	0.0001977	[0.0001567, 0.0002420]
FIFO	Reservoir	0.0001926	[0.0001542, 0.0002340]
Reservoir	FIFO	0.0001833	[0.0001463, 0.0002209]
Reservoir	Reservoir	0.0002709	[0.0002101, 0.0003327]

These gains quantify potential headroom for choosing what to forget. The best deletion is an offline oracle that uses true future errors, so the gain is not evidence that an online learned selector can obtain it. One run's 32-second comparison remains uncertain even though the pooled comparisons are positive.

Continuous replay

Table 3. Validation MSE during irreversible replay with historical capacity 16. The same recent buffer and predictor are used throughout.

Policy	8 s MSE	32 s MSE
FIFO	0.39907	0.47453
Reservoir	0.39987	0.47548
Uniform	0.39910	0.47456

FIFO and Uniform have very similar mean errors; Reservoir is slightly higher at these horizons. These differences are descriptive, with no policy-comparison intervals reported. All replay trajectories respected the memory limit and prevented deleted observations from returning.

Conclusion

The results give partial support to the research premise: historical visual information provides a modest longer-horizon prediction benefit, and deletion choices can change future prediction cost. They establish room for better memory decisions, not a deployed or learned deletion method. Robustness across training runs still needs attention. Action-level prediction, activity changes and recovery remain unevaluated because the exact video-frame and action-label alignment is unresolved.